EXERCISE SHEET 1

GALOIS THEORY

1.1. Consider $\mathbb{Z}_3[y]$, and let $I = (y^3 - y + 1)$ and $J = (y^3 - y - 1)$ be ideals in $\mathbb{Z}_3[y]$.
(a) Why are $\mathbb{Z}_3[y]/I$ and $\mathbb{Z}_3[y]/J$ fields?

(b) What unique form do the elements in both these fields have, respectively?

(c) Let $\theta : \mathbb{Z}_3[y]/I \to \mathbb{Z}_3[y]/J$ be given by $\theta(f(y) + I) \mapsto f(y) + J$ for $f(y)$ a polynomial in $\mathbb{Z}_3[y]$ of degree at most 2. Show that θ is not a field isomorphism.

1.2. Consider the transcendental field extensions $\mathbb{Q}(e)$ and $\mathbb{Q}(\pi)$.

(a) What does an arbitrary element in $\mathbb{Q}(e)$ look like? And in $\mathbb{Q}(\pi)$?

(b) Is $\phi : \mathbb{Q}(e) \to \mathbb{Q}(\pi)$, $\phi(\frac{f(e)}{g(e)}) \mapsto \frac{f(\pi)}{g(\pi)}$ a field isomorphism?

1.3. Suppose that $i : K \to K'$ is a field isomorphism, and let $K(a)$ be an extension of K with m being the minimal polynomial of a over K. Suppose $K'(b)$ is an extension of K' with $i(m)$ being the minimal polynomial of b over K'. Show that the bijection $\mu : K(a) \to K'(b)$ defined by $\mu(a) = b$ and $\mu|_K = i$ is a field homomorphism.

1.4. Let $L : K$ be a finite extension and $f \in K[x]$ an irreducible polynomial of degree $d > 1$. Show that if d and $[L : K]$ are coprime, then f has no roots in L.

1.5. Let $\omega = e^{2\pi i/3}$. Show that $\mathbb{Q}(\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2}) = \mathbb{Q}(\sqrt[3]{2}, \omega)$.